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April 28, 1965

INTERIM FEDERAL SPECIFICATION

ROLLER, HAND (RUBBER ROLLER)

This Interim Federal Specification was developed by General Services Administration, Federal Supply Service, Standardization Division, Washington, D. C. 20407, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope.- This specification covers one type and size of hand roller with a rubber roller as hereinafter specified. The rollers are used in rubber cementing and vulcanizing work.

1.1.1 Federal specification coverage.- Federal specifications do not include all varieties of the commodity indicated by the titles of the specifications, but cover only those generally used by the Federal Government.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and standards.- The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Standards:
Fed. Std. No. 102 - Preservation, Packaging and Packing Levels.
Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 601 - Rubber; Sampling and Testing.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C., 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, and Seattle, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specification:
MIL-H-15424 - Hand Tools; Packaging of.

Military Standards:
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5120

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3. REQUIREMENTS

3.1 Illustration.- Except for dimensional requirements, the illustration shown herein is for the purpose of identification and is not intended to preclude the purchase of hand rollers which otherwise meet the requirements of this specification. The hand rollers shall meet the specified dimensions.

3.2 Material.

3.2.1 Roller.- The roller shall be made of rubber, synthetic rubber, or combination thereof.

3.2.2 Roller core, yoke members, and roller mounting(s).- The roller core, yoke members, and the roller mounting(s) (one or two shafts or studs) shall be of steel. Those portions which form a part of the ball race shall be either through-hardened or case-hardened to such an extent that they are resistant to easy cutting with a file. (Testing with a file to determine that the ball race portions have been hardened will be considered adequate for hardness testing.)

3.2.3 Handle.- The handle shall be made of well-seasoned, straight-grained hardwood such as ash, hickory, or equally hard wood.

3.2.4 Ferrule.- Ferrules shall be of steel or other suitable metal and of rugged design, and when attached, shall act as a support to prevent the handles from splitting or breaking. They shall be pinned, knurled, or otherwise mechanically locked in position on the handle to prevent their becoming loose in service.

3.3 Bearings.- The hand roller shall be equipped with ball bearings at each end of the roller. When assembled, the ball bearings shall provide for free rolling of the roller when hand operated. At the option of the manufacturer, the ball races may be integral portions of the hand roller or they may be parts of the ball bearings.

3.4 Marking.- Each hand roller shall be legibly and permanently marked with the manufacturer's name or trademark of such known character that the source of manufacture can be readily determined.

3.5 Finish and coating.- The wood handle shall be smoothly finished on the hand gripping portion. The exposed surfaces of the wood handle shall be coated with a clear lacquer or varnish so as not to conceal the grain in the wood. The exposed surfaces of the rubber roller shall be smooth.

3.6 Roller.- The roller shall consist of a steel core covered with a rubber roller having a rubber thickness of not less than 11/32 inch.

3.7 Roller hardness.- The durometer hardness value of the rubber roller shall be not less than 42 nor more than 60. The hardness values shall be determined in accordance with test method 3021 in Federal Standard No. 601.

3.8 Size.- The hand roller shall be of the size shown in figure 1. The hand rollers shall be similar to figure 1, except that riveted heads are acceptable.

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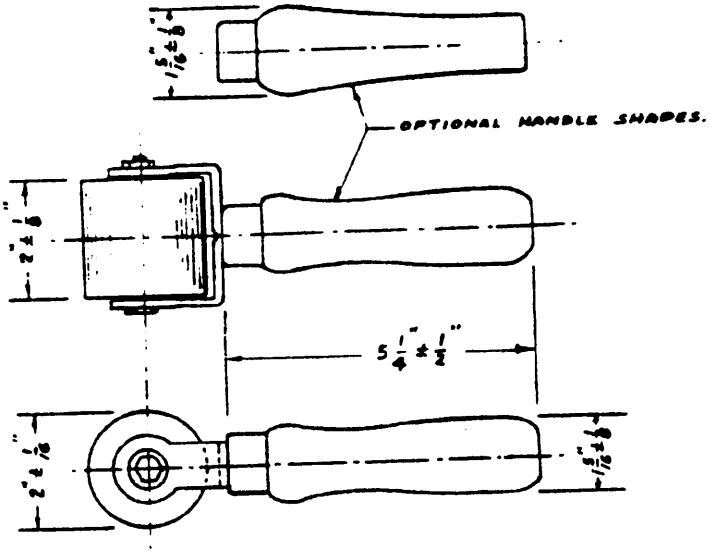


Figure 1. - Roller, hand (rubber roller).

3.9 Workmanship.- Workmanship shall be first class in every respect.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components.- In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures.- Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as stated in table I.

Table I

Sampling Data					
Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	1 ea.	II	4.0	Defects per hundred units	4.3.1
Dimensional examination	1 ea.	S-4	1.0	Defects per hundred units	4.3.2
Testing	1 ea.	S-4	1.0	Percent defective	4.4

4.3 Examination.

4.3.1 Visual examination.- Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking.

4.3.2 Dimensional examination.- Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing.- Each sample unit shall be tested in accordance with 4.4.1.

4.4.1 Hardness of roller.- The Durometer hardness of the rubber on the roller shall be tested in accordance with test method 3021 in Federal Standard 601.

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4.5 Inspection of preparation for delivery.- Inspection to determine compliance with preparation for delivery requirements shall be accomplished in accordance with Specification MIL-H-15424 for Hammer (Handled) Rubber. The inspection level shall be S-4 with an acceptable quality level (AQL) of 1.5 percent defective.

5. PREPARATION FOR DELIVERY

(Civil agencies.- Fed. Std. No. 102 should be referred to for definitions and applications of various levels of packaging protection for supplies and equipment.)

5.1 Preservation and packaging.- See 6.1(d).

5.1.1 Level A.- Level A cleaning, preservation, and packaging shall be in accordance with level A of MIL-H-15424 for Hammers (Handled) Rubber.

5.1.2 Level C.- Preservation and packaging which will afford adequate protection against corrosion, deterioration, and damage during shipment from supply sources to the first receiving activity for immediate use (vendor to user).

5.2 Packing.- See 6.1(d).

5.2.1 Level A.- Level A packing shall be in accordance with level A of MIL-H-15424 for Hammers (Handled) Rubber.

5.2.2 Level B.- Level B packing shall be in accordance with level B of MIL-H-15424 for Hammers (Handled) Rubber.

5.2.3 Level C.- Packing which will afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use.

5.3 Marking.

5.3.1 Civil agencies.- In addition to any special marking specified in the contract or order, each unit and intermediate package and shipping container shall be marked in accordance with Fed. Std. No. 123.

5.3.2 Military.- Interior packages and shipping containers shall be marked in accordance with Military Standard MIL-STD-129.

6. NOTES

6.1 Ordering data.- Purchasers should exercise any desired option offered herein and procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation, packaging and packing required (see 5.1 and 5.2).
(c) Special marking requirements (see 5.3.1).

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